Check for updates

JGR Planets

RESEARCH ARTICLE
10.1029/2024JE008615

# Radar Attenuation for Subsurface Sounding on Enceladus: Effects of a Thermally Insulating Porous Ice Layer

**William Byrne[1,2], Ana-Catalina Plesa[1], Tina Rückriemen-Bez[1], Hauke Hussmann[1], and Andreas Benedikter[3]**

[1]Institute of Planetary Research, German Aerospace Center (DLR), Berlin, Germany, [2]University of Maryland, College Park, MD, USA, [3]Microwaves and Radar Institute, German Aerospace Center (DLR), Weßling, Germany

**Key Points:**

- We calculate the two-way radar attenuation on Enceladus considering a porous thermally insulating surface layer
- For regions covered by a thick insulating porous surface layer the detection of the ice-ocean interface is unlikely
- For the same regions the high subsurface temperatures increase the likelihood that shallow brines are present and can be detected by radar

**Supporting Information:**

Supporting Information may be found in the online version of this article.

**Correspondence to:**

W. Byrne,
William.Byrne@dlr.de

**Citation:**

Byrne, W., Plesa, A.-C., Rückriemen-Bez, T., Hussmann, H., & Benedikter, A. (2024). Radar attenuation for subsurface sounding on enceladus: Effects of a thermally insulating porous ice layer. *Journal of Geophysical Research: Planets*, *129*, e2024JE008615. https://doi.org/10.1029/2024JE008615

Received 16 SEP 2024
Accepted 28 NOV 2024

© 2024. The Author(s).
This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

**Abstract** Saturn's moon Enceladus is thought to possess habitable conditions due to the presence of a global ocean, potential water-rock interactions at the ocean's base, and its ocean composition derived from sampling active plumes at the south pole. The thin (<35 km) and cold ice shell of this small moon in the outer solar system is expected to allow the direct detection of the ice-ocean interface using low-frequency radar sounder instruments. Here we investigate the two-way radar attenuation in Enceladus' ice shell, focusing on the effect of a porous icy layer generated by Enceladus' jet activity. Our results show that in regions with a thin (or absent) porous surface layer, total ice shell penetration is possible. In regions covered by thick and strongly insulating porous surface layers, however, as little as 2% of the ice shell can be penetrated. Nevertheless, a thick porous surface layer leads to high subsurface temperatures promoting the formation of brines at shallow depth that can be detected by future radar measurements.

**Plain Language Summary** Saturn's moon Enceladus is a prime target for planetary exploration and for the search of habitable conditions beyond Earth. Beneath its icy surface, this small moon is thought to harbor a global ocean, presumably sampled by active water jets which have been observed at Enceladus' south pole. Moreover, shallow brines may exist within the ice shell. The detection of subsurface water reservoirs (either the ocean or shallow brines) that can be achieved by radar is fundamental in characterizing Enceladus' subsurface environment and its habitability potential. In this study we calculate the attenuation of radar signals through the ice shell in the presence of snow deposits that are believed to exist on the surface of Enceladus due to its water jets activity. While we do identify some scenarios in which the radar is able to reach the ice-ocean interface, we note that this interface may not be reached by a radar sounding instrument in regions covered by thick snow deposits that act as a blanket and keep the subsurface warm. However, in these regions, due to the high subsurface temperatures, shallow water bodies are likely to exist and could be detected by future radar observations.

## 1. Introduction

Saturn's moon, Enceladus is considered a priority target for future planetary missions due to its high astrobiological potential (Cable et al., 2021; Choblet et al., 2021). Water jets presumably originating from a subsurface ocean have been observed at the south pole of Enceladus by NASA's Cassini mission (e.g., Hansen et al., 2006; Porco et al., 2006). Their analysis provides a direct window into the ocean composition (Postberg et al., 2009) which, in turn, can help to understand the nature and amount of impurities that may exist within the ice shell. The most direct way to investigate the ice shell structure would be with future radar measurements. Radar instruments are widely used to characterize polar ice sheets on the Earth (e.g., Schroeder et al., 2020) and Mars (Phillips et al., 2008; Plaut et al., 2007), and are part of the instrument suites of JUICE and Europa Clipper which will be used to investigate the ice shells of icy moons in the Jupiter system (Blankenship et al., 2024; Bruzzone et al., 2013).

The thickness of Enceladus' ice shell is an important parameter when considering future exploration by radar measurements. It is thought to vary gradually depending on latitude and longitude (Čadek et al., 2019; Hemingway & Mittal, 2019) being thinnest at the south-polar region and thickest at the equatorial sub- and anti-Saturn points. Depending on the chosen shape model (Nimmo et al., 2011; Tajeddine et al., 2017) and assumed density contrast between the ice shell and the ocean, ice shell thickness models suggest strong variations, with values as low as 5 km at the south pole and as high as 35 km in equatorial regions, or a more homogeneous ice shell thickness distribution with less than 15 km differences (Hemingway & Mittal, 2019).

21699100, 2024, 12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024JE008615 by Dtsch Zentrum F. Luft-U. Raum Fahrt In D. Helmholtz Gemein., Wiley Online Library on [01/10/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Enceladus' jet activity likely leads to the formation of a highly porous surface layer at the top of the ice shell due to the material fall-back from the plume eruptions at the south pole (Kempf et al., 2010). If present, such a porous surface layer could be stable on most of Enceladus' surface due to the low temperatures (Gundlach et al., 2018). The thickness of this porous surface layer and its distribution are poorly constrained, but local thicknesses of up to 700 m have been reported from the analysis of pit chains on the surface of Enceladus (Martin et al., 2023). Such a thick porous surface layer may strongly affect the thermal state of the ice shell, leading to higher signal attenuation (Souček et al., 2023), but its effects have not been investigated so far.

Another aspect that can affect the subsurface temperature and hence the two-way attenuation is the way heat is currently transported through Enceladus's ice shell. Heat transport through the ice shell could occur either by conduction or by convection and can affect the thermal state of the ice shell with typically lower temperatures for a conductive scenario. Here we focus on a purely conductive thermal scenario. This scenario is very likely given the thin ice shell (<35 km) and high viscosity at the conditions relevant for Enceladus's ice shell (Barr & McKinnon, 2007). High viscosities are expected based on the grain sizes observed in polar ice sheets on the Earth (between 1 and 5 mm, Montagnat & Duval, 2000) that are so far the only analogs we have for the icy shells in the outer solar system.

## 2. Methods

### 2.1. Thermal State of Enceladus's Ice Shell

To study how ice shell properties impact radar attenuation, we use 1D steady-state conductive thermal models of Enceladus's interior. The model is solved numerically using the stationary heat-conduction equation for a spherical shell with an outer radius $R_{top}$ of 252 km and an inner radius $R_{bot}$, which is varied to account for ice shell thickness variations on Enceladus. A semi-analytical solution for the temperature profile is provided in the Supporting Information S1, in Text S1. The steady-state heat conduction equation is given below:

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(kr^2\frac{\partial T}{\partial r}\right)=0, \tag{1}$$

where $r$ is the radius, $k$ is the thermal conductivity of ice, and $T$ is the temperature. The thermal conductivity of ice is temperature-dependent and follows the parametrization of Petrenko and Whitworth (1999):

$$k(T)=\frac{651}{T}. \tag{2}$$

We note that a more recent parametrization for the ice thermal conductivity has been derived by Wolfenbarger et al. (2021). Although this new parametrization has a larger range of validity for temperatures between 30 and 273.15 K, our tests indicate that the results are similar to those obtained using the parametrization of Petrenko and Whitworth (1999) for which the range of validity is given for temperatures >80 K (Wolfenbarger et al., 2021). A comparison of the thermal conductivity and temperature profiles for these two parametrizations is shown in Figure S2.1 in Supporting Information S1.

We solve Equation 1 numerically and apply boundary conditions using the surface temperature of Enceladus at the top and the melting temperature of water-ice at the ice-ocean interface. For the surface temperature we use an average value of 60 K representative for most of Enceladus' surface, but note that the surface temperature can have higher local values in the south polar region, near the tiger stripes. To account for this, we additionally test the effect of a higher surface temperature of 85 K, which is representative of the average local temperature increase in this region (Spencer & Grinspoon, 2007). We test the effects of higher surface temperatures in additional simulations that are presented in Figure S2.1 in Supporting Information S1 and in the Discussion section.

In contrast to previous studies that calculated the two-way radar attenuation in icy shells (Aglyamov et al., 2017; Chyba et al., 1998; Kalousová et al., 2017; Moore & Schubert, 2000; Souček et al., 2023), we consider the effects of a porous ice layer specifically as a thermal insulator. This porous ice layer may considerably affect the thermal state of the ice shell due to its low thermal conductivity. Thermal conductivity values have been measured for porous icy layers in laboratory experiments (Seiferlin et al., 1996) and derived from thermophysical modeling (Ferrari et al., 2021), yielding values between 0.025 and 0.001 $Wm^{-1}K^{-1}$. Therefore, in order to cover a wide

parameter space, we vary the thermal conductivity of this porous surface layer between 0.001 and 0.1 $Wm^{-1}K^{-1}$. Note, that thermal conductivity is assumed to be constant in the porous layer in contrast to a temperature-dependent thermal conductivity in the ice shell below (see Equation 2).

### 2.2. Radar Attenuation

Radar attenuation in ice is caused by dielectric absorption losses, interface transmission losses, as well as scattering losses within the volume. A common assumption in low-frequency radar sounding is that scattering losses are negligible due to the long wavelengths with respect to the scattering structures (e.g., ice grains) within the ice and the porous surface layer. Transmission losses occur at dielectric interfaces formed between: (a) vacuum and the porous surface layer, (b) the porous surface layer and the ice, as well as (c) ice-internal dielectric horizons. The dielectric contrast of these interfaces is expected to result in transmission losses that only contribute marginally (e.g., at most a few dB) to the overall attenuation budget (Grima et al., 2024). Hence, attenuation purely driven by absorption losses is analyzed here. Following the work of Kalousová et al. (2017) and Souček et al. (2023), we calculate the two-way attenuation along the propagation path at depth $d$, which was subdivided into $N$ layers, as follows:

$$A_2(d) = \sum_{i=0}^{N} 2A_1(d_i)\Delta d \tag{3}$$

where $d_0$ represents the surface layer, $d_N$ is the final layer, $A_1(d_i)$ is the one-way attenuation rate at depth $d_i$, and $\Delta d$ the thickness of each layer used in calculating $A_1$.

The one-way attenuation rate $A_1$ (in dB/km) of a radar signal in ice can be approximated (Matsuoka et al., 2012) as proportional to the high frequency (HF) electrical conductivity ($\sigma$) in units of μS/m as follows:

$$A_1 \approx 0.914\sigma. \tag{4}$$

The HF electrical conductivity of ice in terrestrial sounding experiments depends on the ice temperature along with the presence of chemical impurities soluble within the ice crystal lattice, and can be written as follows (Corr et al., 1993):

$$\sigma = \sum_{i=0}^{1} \sigma_i^0 C_i \exp\left[-\frac{E_i}{k_B}\left(\frac{1}{T} - \frac{1}{T_r}\right)\right]. \tag{5}$$

Each summand, representative of a chemical impurity, requires the material molar electrical conductivity $\sigma_i$ at reference temperature $T_r = 251$ K, molar concentration $C_i$, activation energy $E_i$, local ice temperature $T$, and the Boltzmann constant $k_B$. In this study we use the same values as in Table 1 of Souček et al. (2023). The index 0 indicates a scenario of a pure water-ice shell, while $i = 1$ adds the effects of the maximum theoretical chloride concentration in pure water-ice.

We follow the approach of Kalousová et al. (2017) and Souček et al. (2023) when distinguishing end members in terms of chemical impurities. Similar to Souček et al. (2023), we use the term "low-loss" to indicate that the attenuation model considers only pure water ice, and "high-loss" to characterize a model considering pure water ice doped with 300 μM of chloride. Although carbonate salts, chloride, silica, and ammonia/ammonium have been detected within plume material (Fox-Powell & Cousins, 2021; Glein et al., 2018), Souček et al. (2023) argued that chloride will be the only significant contributor to the two-way attenuation of a radar sounder in terms of HF electrical conductivity. We assume the same attenuation limit of 100 dB (Kalousová et al., 2017), which we note is a representative value that in practice depends on mission modifications, instrument design, and environmental conditions. To quantify the effects of the attenuation limit on our results, we test values of 70 and 130 dB (Figures S4.1–S4.2 in Supporting Information S1), and refer to the Discussion section for more details.



21699100, 2024, 12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024JE008615 by Dtsch Zentrum F. Luft-U. Raum Fahrt In D. Helmholtz Gemein., Wiley Online Library on [01/10/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

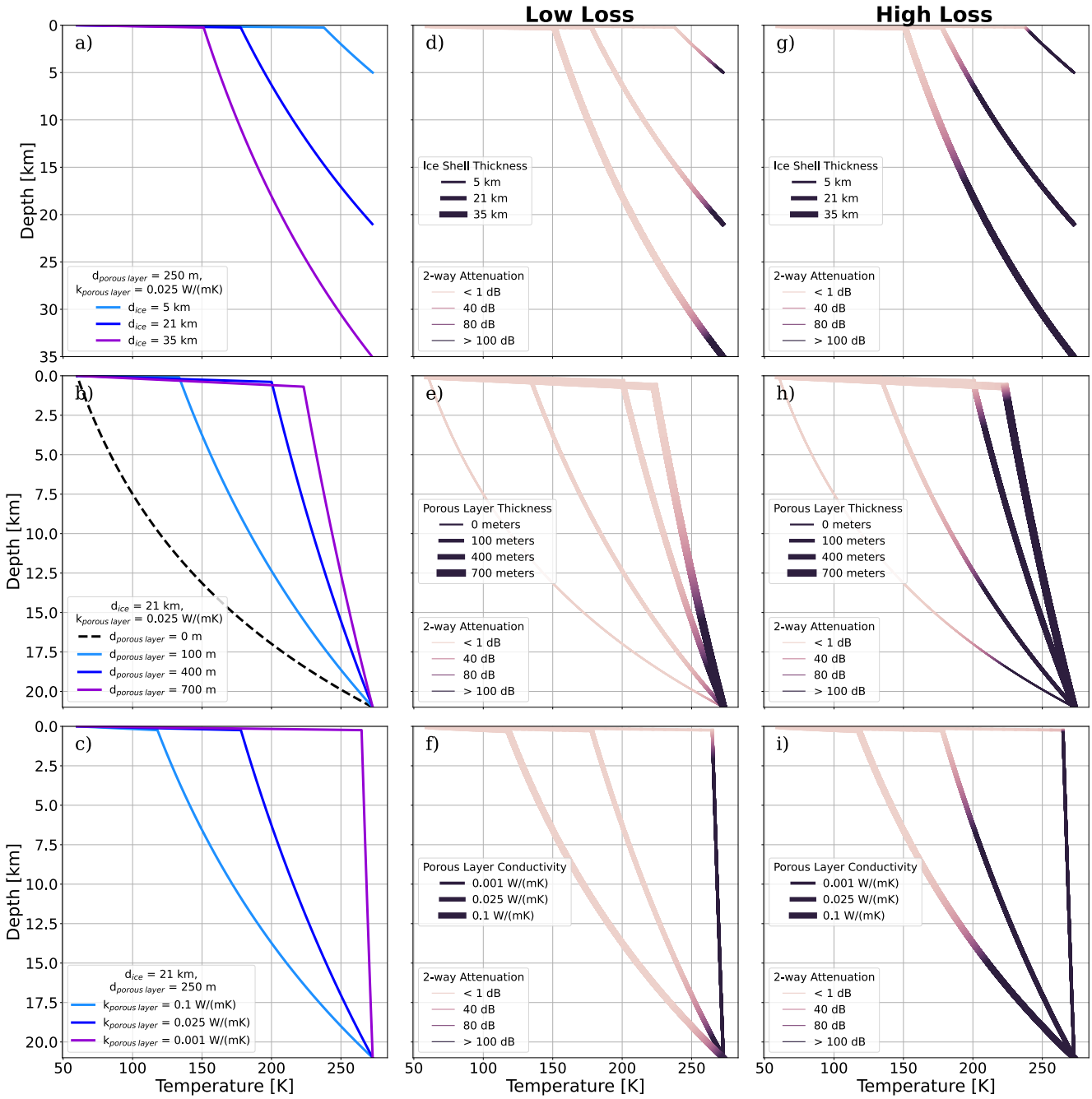



**Figure 1.** The two-way attenuation ($A_2$) for various (a) ice shell thicknesses (5–35 km), (b) porous surface layer thicknesses (0–700 m), and (c) porous surface layer thermal conductivities (0.1–0.001 $Wm^{-1}K^{-1}$). Panels (d)–(f) show the low-loss scenarios, whereas panels (g)–(i) show the high-loss cases.

## 3. Results

First, we investigate the difference between the "high" and "low" loss attenuation models in multiple 1-dimensional scenarios of Enceladus' internal structure. To this end, we calculate temperature profiles and corresponding two-way attenuation for various ice shell thicknesses, thicknesses of the porous surface layer, and porous surface layer thermal conductivities (Figure 1).

Following Kalousová et al. (2017) and Souček et al. (2023), we define a penetration depth, which is the depth at which the two-way attenuation reaches a 100 dB. The relative penetration depth is the penetration depth divided



21699100, 2024, 12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024JE008615 by Dtsch Zentrum F. Luft-U. Raum Fahrt In D. Helmholtz Gemein., Wiley Online Library on [01/10/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

21699100, 2024, 12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024JE008615 by Dtsch Zentrum F. Luft-U. Raum Fahrt In D. Helmholtz Gemein., Wiley Online Library on [01/10/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

by the total ice shell thickness of the respective scenario. The relative penetration depth strongly depends on temperature, as the latter affects the thermal conductivity (Equation 5). For a thermally conductive ice shell, the temperature itself depends on the thickness of the ice shell, and to a first order on the presence of a porous surface layer. The porous surface layer leads to a high thermal gradient near the surface due to its insulating effect caused by its low thermal conductivity (Figures 1a–1c). The relative penetration depth in a low-loss scenario is 91%, 94%, and 95%, for ice shell thicknesses of 5 km (representative for the south polar region), 21 km (average ice shell thickness), and 35 km (thickest ice shell in equatorial regions), respectively, based on values from Hemingway and Mittal (2019), as illustrated in Figure 1d. The penetration depths relative to the total depth in the high-loss case are 16%, 33%, and 48%, for the same ice shell thicknesses of 5 km, 21 km, and 35 km, respectively (Figure 1g).

We can clearly observe in Figures 1e and 1h that the attenuation dramatically increases with increasing porous surface layer thickness. This can be attributed to the insulating effect of the porous surface layer, an effect that becomes more pronounced with increasing porous surface layer thickness. As observed in Figure 1e, the difference between the temperature profiles with and without a porous surface layer of 700 m is as large as 150 K at about 2.5 km depth. The low-loss scenario with no porous surface layer only reaches 53.4 dB at the ice-ocean interface, which was set here to 21 km. In the low-loss scenario, the presence of a 100 m porous surface layer increases the two-way attenuation to 100 dB or higher at 99% of the ice shell depth. The relative penetration depth associated with the 100 dB limit is 88% for a 400 m and 74% for a 700 m porous surface layer. For the high-loss scenario (Figure 1h), the radar attenuation increases to more than 100 dB before reaching the ice-ocean interface even when no porous surface layer is considered. Thus, the relative penetration depth is 88% of the total ice shell for the no porous surface layer case, 64% for the 100 m porous surface layer case, 16% for the 400 m case, and 8% for the 700 m porous surface layer.

The thermal conductivity of the porous surface layer can significantly impact the thermal profile within the ice shell. A highly insulating porous surface layer (i.e., thermal conductivity of 0.001 $Wm^{-1}K^{-1}$) with a thickness of 250 m, leads to a warm ice shell, only a few degrees colder than the ice-ocean interface (Figure 1c). In the low-loss scenario, the two-way attenuation for the case with the highest thermal conductivity of the porous surface layer (i.e., 0.1 $Wm^{-1}K^{-1}$) reaches 95 dB at the ice-ocean interface. For a thermal conductivity of 0.025 $Wm^{-1}K^{-1}$ of the porous surface layer, the two-way attenuation reaches the 100 dB limit at about ~80% of the ice shell in the low loss case, while in the high-loss case the 100 dB are reached after only about ~30% of the ice shell. In the case with the most insulating porous surface layer (i.e., porous surface layer thermal conductivity of 0.001 $Wm^{-1}K^{-1}$), the two-way attenuation reaches 100 dB before 10% of the ice shell is penetrated in both the high and low-loss scenarios (Figures 1f and 1i).

In Figure 2, we explore a large set of parameter combinations. Each cell represents a single simulation (equivalent to one line graph in Figure 1). The value inside each cell represents the relative penetration depth. The simulations cover ice shell thicknesses of 5 km, 21 km, and 35 km (top, middle, and bottom row, respectively), as well as the low-loss and high-loss attenuation models (left and right column, respectively). None of the high-loss models reach 100% relative depth penetration, even with a 5 km ice shell with no porous surface layer. Most simulations using a highly insulating porous surface layer with a thermal conductivity of 0.001 $Wm^{-1}K^{-1}$ reach less than 10% relative penetration depth. Of the total of 90 simulations, 16 models penetrate the whole ice shell, (7 models for a 5-km-thick ice shell, 5 models for a 21-km-thick ice shell, and 4 models for a 35-km-thick ice shell). These 16 models generally have thin porous surface layers or a high thermal conductivity of the porous surface layer or both.

As expected, every combination of parameters has a lower relative penetration depth in the high-loss attenuation model compared to low-loss. Second, the relative penetration depth decreases with increasing porous surface layer thickness and decreasing thermal conductivity. However, a trend reversal occurs (i.e., penetration depth increases) in highly insulating porous surface layers (apart from the 35 km low-loss scenarios). Such a reversal only occurs in scenarios with both a highly conductive and very deep porous surface layer. This behavior can be explained by the fact that with increasing porous surface layer thickness, the thermal state of the ice shell starts to approach that of a conductive ice shell characterized by a single thermal conductivity. If the porous surface layer makes up the entire ice shell, the steady-state thermal state will be identical to that of a homogeneous ice shell with a constant thermal conductivity of pure water-ice.

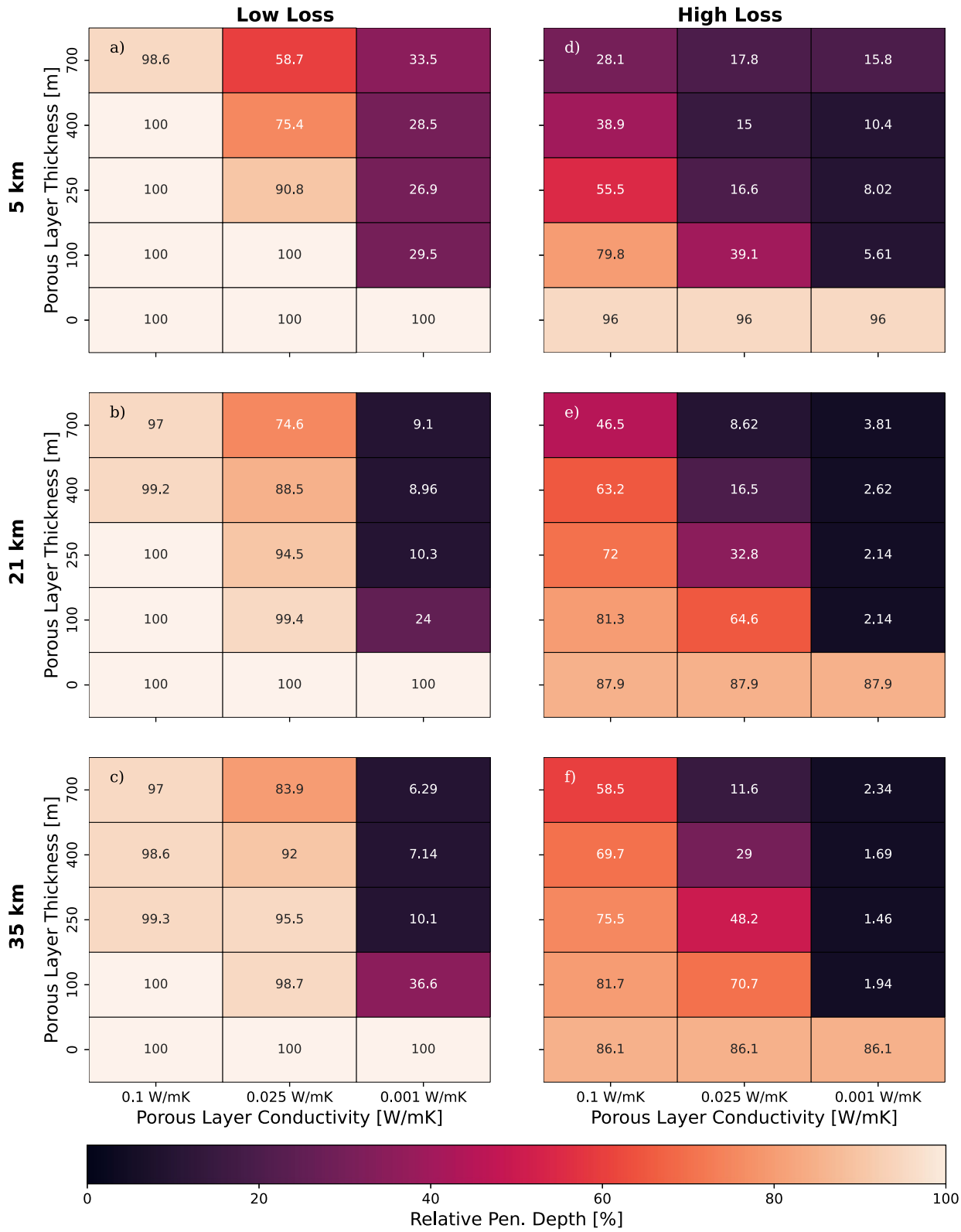



**Figure 2.** The relative penetration depth of a radar signal for ice shell thicknesses of 5 km (a), (d), 21 km (b), (e), and 35 km (c), (f) and various porous surface layer thicknesses and thermal conductivities for low-loss (left column) and high-loss (right column) scenarios.



21699100, 2024, 12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024JE008615 by Dtsch Zentrum F. Luft-U. Raum Fahrt In D. Helmholtz Gemein., Wiley Online Library on [01/10/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

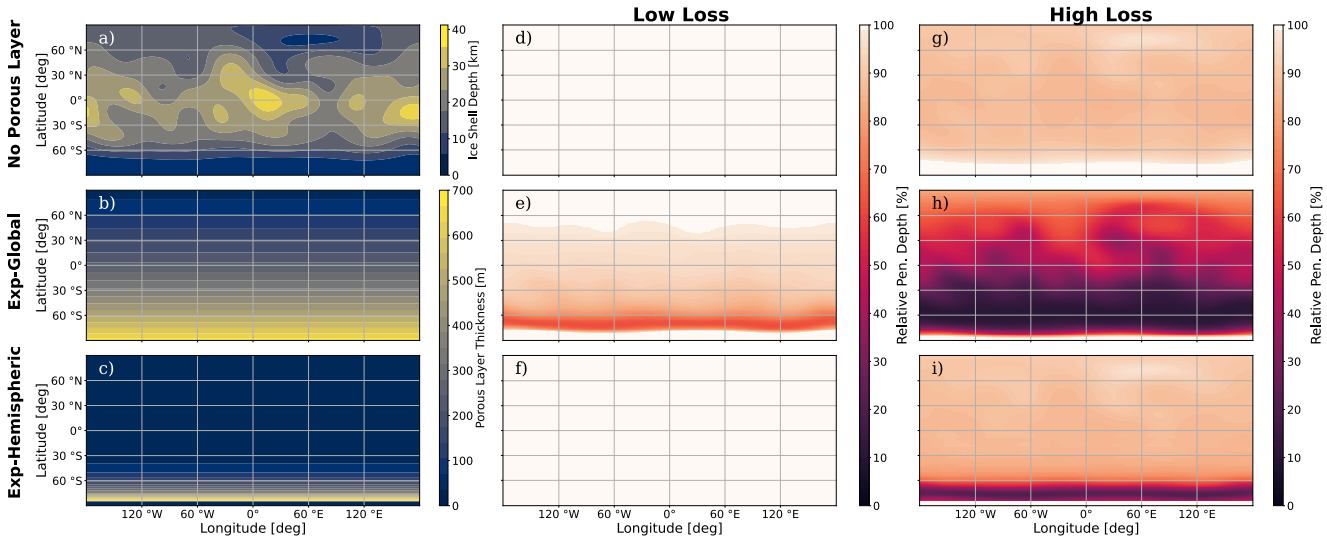



**Figure 3.** Maps of the (a) ice shell thickness (Hemingway & Mittal, 2019) and porous surface layer distribution for (b) the "exponential-global (Exp-Global) scenario" and (c) the "exponential-hemispheric scenario" (Exp-Hemispheric) (see text). The radar penetration depth relative to the ice shell thickness in panel (a) is shown for (d) no porous surface layer, (e) exponential-global and (f) exponential-hemispheric distribution of the porous surface layer. Panels (g), (h), and (i) show the high-loss cases.

Beyond one-dimensional analysis, we generate attenuation maps (Figure 3a) using a global ice shell thickness map from Hemingway and Mittal (2019). Additional attenuation maps using a different ice shell thickness, shape model and ice-ocean density contrast are available in Supporting Information S1, Text S5.

The attenuation was calculated by treating each latitude and longitude point as a single temperature profile using the corresponding ice shell thickness at the respective location. We constructed two distribution models for the porous surface layer under the assumption that the thickness of the porous surface layer varies only with latitude and: (a) exponentially decreases from a 700-m-thick layer at the south pole to a 10-m-thick layer at the north pole of Enceladus, which we denote as the "exponential-global distribution" (Figure 3b) while (b) the porous surface layer is absent within the 5° of the south pole and then exponentially decreases from 255 to 0 m at equator (Figure 3c), which we will call "exponential-hemispheric distribution." In all cases the conductivity of the porous surface layer is 0.025 $Wm^{-1}K^{-1}$.

In the low-loss attenuation maps (Figures 3d–3f), a majority of the surface has more than 90% relative penetration depth, with both the no-porous layer and the exponential-hemispheric distribution cases having 100% global radar penetration down to the ice-ocean interface. In all three high-loss scenarios, we observe a small band of 100% relative penetration depth roughly 5° in latitude away from the south pole. The thin ice shell at the south pole allows for the signal to reach the ice-ocean interface even in the presence of a thick porous surface layer (i.e., 700 m in the exponential-global distribution case).

We note that in all cases besides the high-loss exponential-global distribution case, the penetration depth reaches more than 80% of the ice shell for most locations.

In Figure 4, we calculate the two-way attenuation at the eutectic interfaces of ammonium chloride and ammonia following Souček et al. (2023) for the exponential-hemispheric distribution of the porous surface layer (i.e., exponential decrease from 255 m at 85°S to 0 m at equator) and assuming a thermal conductivity of the porous surface layer of 0.025 $Wm^{-1}K^{-1}$.

The eutectic interface of ammonium chloride ($NH_4Cl$) is defined by the 257.79 K isotherm, while the one for ammonia ($NH_3$) by the 175.45 K isotherm (Marion et al., 2012). Panels a and b of Figure 4 show the depths at which the eutectic interface of $NH_3$ and $NH_4Cl$, respectively, are reached. Figure 4c and d show the two-way attenuation for signals traveling to the eutectic interfaces using the low-loss attenuation models, while the values obtained for the high-loss cases are shown in Figures 4e and 4f. Additional maps including different ice



21699100, 2024, 12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024JE008615 by Dtsch Zentrum F. Luft-U. Raum Fahrt In D. Helmholtz Gemein., Wiley Online Library on [01/10/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

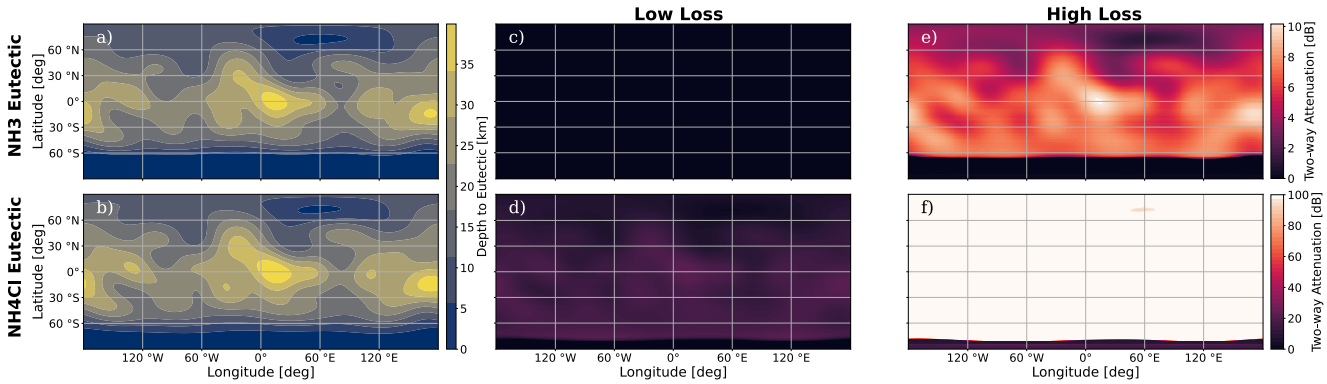



**Figure 4.** Maps showing the depth of: (a) ammonia ($NH_3$) and (b) ammonium chloride ($NH_4Cl$) eutectic interface for the exponential-hemispheric distribution of the porous surface layer. Panels (c) and (d) indicate the corresponding two-way attenuation values for the low-loss case at the eutectic interface of $NH_3$ and $NH_4Cl$, respectively. Panels (e) and (f) are similar to (c) and (d), but for the high-loss scenario. Note that the two-way attenuation colorbar scale is different between upper and lower row.

shell thicknesses and porous surface layer distribution are available in the SI (Text S6 in Supporting Information S1).

The difference between the average depth of the ice-ocean interface and the average depth of the $NH_4Cl$ eutectic interface is only 700 m. This is due to the fact that the temperature of the $NH_4Cl$ eutectic interface is only 15 K lower than the melting temperature of water-ice at the base of the ice shell. In contrast, the difference between the average depth of the eutectic interface of ammonia and that of the ice-ocean interface is 5 km, as $NH_3$ has a much lower eutectic temperature of only 175.45 K. Given the low eutectic temperature of $NH_3$, the two-way attenuation for the low-loss case (Figure 4c) does not exceed 0.1 dB. For the eutectic interface of $NH_4Cl$, the two-way attenuation shows an average value 13.7 dB in the low-loss scenario (Figure 4d). In the high-loss attenuation case, the two-way attenuation at the eutectic interface of $NH_3$ is only ~5 dB (Figure 4e), while at the eutectic interface of $NH_4Cl$ the average two-way attenuation is about 400 dB (Figure 4f). Note that the scale is cut at 100 dB because this study assumes 100 dB as a detection limit.

## 4. Discussion

On Enceladus, the surface temperature reaches about 85 K on average at the south polar region, with values higher than 145 K in localized areas close to the tiger stripes. Since we are interested in regional rather than small scale variations, we tested the effects of a surface temperature of 85 K, as suggested for the south pole of Enceladus, on our calculated radar attenuation values (Figure S3.1 in Supporting Information S1). A higher surface temperature shifts the ice shell temperature to higher values and leads to a smaller relative penetration depth. However the differences between using a surface temperature of 85 K compared to 60 K are only of the order of 2%–4%.

Similar to other studies (Kalousová et al., 2017; Souček et al., 2023), we assume a threshold of 100 dB for the two-way radar attenuation, above which the radar instrument would no longer be able to detect subsurface interfaces. However, we note that this threshold depends on the instrument design, operation scenarios, as well as surface and subsurface characteristics (see e.g., Benedikter et al., 2024; Kalousová et al., 2017, for a detailed discussion). Our tests show that varying the two-way attenuation limit between 70 and 130 dB would not significantly impact the results presented in Figure 2. In these cases we observe a difference to the relative penetration depth shown in Figure 2 of up to only 10% (see Figures S4.1 and S4.2 in Supporting Information S1).

We note that our study assumes simple distributions of a porous surface layer on Enceladus to give a first order overview of the potential detection of the ice-ocean interface via radar sounding. While our results show that the ice-ocean interface will be challenging to detect in particular in regions covered by thick and highly thermal insulating porous surface layers, we note that large uncertainties exist for the global spatial distribution of such a porous surface layer on Enceladus. Although models of the distribution of a porous surface layer on Enceladus exist (Kempf et al., 2010; Southworth et al., 2019), they typically investigate only the generation of such a layer through plume deposition. Such models indicate that tectonized regions on the leading and trailing hemispheres



21699100, 2024, 12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024JE008615 by Dtsch Zentrum F. Luft-U. Raum Fahrt In D. Helmholtz Gemein., Wiley Online Library on [01/10/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

21699100, 2024, 12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024JE008615 by Dtsch Zentrum F. Luft-U. Raum Fahrt In D. Helmholtz Gemein., Wiley Online Library on [01/10/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

receive the least amount of material deposition, but the plume eruption style (i.e., curtain-like vs. jet-like) affects the amount of snow deposits in the northern hemisphere (Southworth et al., 2019). The predicted distribution of a material deposition from models of jet style eruptions seem to be consistent with the distribution of pit chains on the surface of Enceladus. The latter, however, can only give a lower bound for the thickness of the porous surface layer, as the pits do not necessarily need to reach the base of the layer (Martin et al., 2023).

Another important factor that will affect the thermal state of the ice shell and hence the two-way attenuation is the thermal conductivity of the porous surface layer. A recent study by Jabaud et al. (2024) investigated the mechanical properties of powder-like materials similar to the expected snow deposits on Enceladus that are also suggested to exist to some extent on Europa. This study concluded that on Enceladus, the lower gravity leads to higher cohesion and the formation of loose highly porous material. A high porosity is directly linked to a low thermal conductivity and thus highly insulating behavior. As shown in this study, an insulating porous surface layer would increase the ice shell temperature such that the resulting temperature contrast between the base of the porous surface layer and the ocean may be only 10s of degrees K. Such high average temperatures could result in the formation of shallow brines that could provide a radar-detectable source (Souček et al., 2023). Our models show that even in the presence of thick porous surface layers, the two-way radar attenuation remains below 25 dB for eutectic interfaces of $NH_3$ and $NH_4Cl$ in the low-loss scenario. In the high-loss cases, only the $NH_4Cl$ interface lies above the 100 dB threshold for most of the ice shell apart from within about 5° from the south pole. As these eutectic interfaces are defined by isotherms, their detection would provide valuable information about the thermal state of the ice shell and may help reconstructing not only the ice shell thickness (see discussion in Souček et al., 2023) but also the thickness of the porous surface layer that strongly affects the temperature in the ice shell and hence the depth of these interfaces. Moreover, we note that future radar measurements could be used to characterize the thickness and distribution of the porous surface layer (Scanlan et al., 2022).

This study has focused on the effect of a porous surface layer on two-way attenuation in a conductive ice shell. Future studies will test how such a porous surface layer, its thickness, and spatial distribution affects the ice shell dynamics and in particular the two-way radar attenuation for hot upwellings and cold downwellings. Depending on the ice shell temperature, cold downwellings could represent locations where the radar attenuation is low enough such that the ice-ocean interface may be reached, as previously discussed for Europa (Kalousová et al., 2017).

## 5. Conclusions

In this study we showed that a porous surface layer built up by the deposition of material from Enceladus's south pole plume or generated by impacts has a first order effect on the two-way radar attenuation. This porous surface layer acts as an insulation layer leading to high subsurface temperatures and thus two-way attenuation. Through systematic parameter variations, we provide an extensive overview of the effects of a porous surface layer on the ability to reach the ice-ocean as well as other interfaces within the ice shell that are thought to exist based on the inferred ice shell chemistry.

Our investigations show that radar sounder instruments may penetrate anywhere from a few hundred meters to the entire ice shell of Enceladus. Nevertheless, those scenarios that prevent radar signals from reaching the ice ocean interface can provide valuable information about the ice shell. Identifying aspects of the Enceladus system that could prevent ocean detection may prove fundamental in further characterizing icy worlds. Valuable insights may be gained, such as the existence of near-surface liquid reservoirs, which are very likely for cases with high salt content and a porous surface layer on top that keeps the interior comparatively warm. Such brine reservoirs are considered niches for life, and their potential to be directly accessible by future missions is much greater compared to that of the subsurface oceans (Wolfenbarger et al., 2022).

## Data Availability Statement

Additional calculations including different ice shell thicknesses are available in Supporting Information S1. All data used in the figures, the code necessary to reproduce the calculations, and the Supporting Information are available on Zenodo (Byrne et al., 2024).

**Acknowledgments**
ACP and TRB acknowledge funding from the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation)—Project-ID 460819306. AB and HH were supported in part by the European Space Agency under Contract 4000141148/23/NL/KK/ov. We thank Douglas Hemingway for providing the ice thickness maps from Hemingway and Mittal (2019). We thank Gregor Steinbrügge and Dustin Schroeder for insightful discussions on radar attenuation on icy moons. Finally, we thank Natalie Wolfenbarger of Stanford's Radio Glaciology lab for her rewording suggestions that provided much-needed narrative clarity. We thank Gareth Morgan and an anonymous reviewer for their insightful comments that helped improve an earlier versions of this manuscript. We also thank the editor Amanda Hendrix for handling our paper. Open Access funding enabled and organized by Projekt DEAL.

## References

Aglyamov, Y., Schroeder, D. M., & Vance, S. D. (2017). Bright prospects for radar detection of Europa's ocean. *Icarus*, *281*, 334–337. https://doi.org/10.1016/j.icarus.2016.08.014

Barr, A. C., & McKinnon, W. B. (2007). Convection in Enceladus' ice shell: Conditions for initiation. *Geophysical Research Letters*, *34*(9). https://doi.org/10.1029/2006gl028799

Benedikter, A., Matar, J., Masaki, N., Otto, T., Hussmann, H., Perihar, T., et al. (2024). A fractionated radar sounder concept for subsurface exploration of Saturn's icy moon Enceladus. In *Eusar 2024* (p. a261).

Blankenship, D. D., Moussessian, A., Chapin, E., Young, D. A., Wesley Patterson, G., Plaut, J. J., et al. (2024). Radar for Europa assessment and sounding: Ocean to near-surface (reason). *Space Science Reviews*, *220*(5), 51. https://doi.org/10.1007/s11214-024-01072-3

Bruzzone, L., Plaut, J., Alberti, G., Blankenship, D., Bovolo, F., Campbell, B., et al. (2013). RIME: Radar for icy moon exploration. In *2013 IEEE international geoscience and remote sensing symposium-IGARSS* (pp. 3907–3910).

Byrne, W., Plesa, A.-C., Rückriemen-Bez, T., Hussmann, H., & Benedikter, A. (2024). Radar attenuation for subsurface sounding on Enceladus: Effects of a porous ice layer [Dataset]. *Zenodo*. https://doi.org/10.5281/zenodo.10932226

Cable, M. L., Porco, C., Glein, C. R., German, C. R., MacKenzie, S. M., Neveu, M., et al. (2021). The science case for a return to Enceladus. *The Planetary Science Journal*, *2*(4), 132. https://doi.org/10.3847/psj/abfb7a

Čadek, O., Souček, O., Běhounková, M., Choblet, G., Tobie, G., & Hron, J. (2019). Long-term stability of Enceladus' uneven ice shell. *Icarus*, *319*, 476–484. https://doi.org/10.1016/j.icarus.2018.10.003

Choblet, G., Tobie, G., Buch, A., Čadek, O., Barge, L. M., Běhounková, M., et al. (2021). Enceladus as a potential oasis for life: Science goals and investigations for future explorations. *Experimental Astronomy*, *54*(2–3), 1–39. https://doi.org/10.1007/s10686-021-09808-7

Chyba, C. F., Ostro, S. J., & Edwards, B. C. (1998). Radar detectability of a subsurface ocean on Europa. *Icarus*, *134*(2), 292–302. https://doi.org/10.1006/icar.1998.5961

Corr, H., Moore, J. C., & Nicholls, K. W. (1993). Radar absorption due to impurities in Antarctic ice. *Geophysical Research Letters*, *20*(11), 1071–1074. https://doi.org/10.1029/93gl01395

Ferrari, C., Lucas, A., & Jacquemoud, S. (2021). The thermal emission of Saturn's icy moons-Effects of topography and regolith properties. *Astronomy & Astrophysics*, *655*, A8. https://doi.org/10.1051/0004-6361/202141223

Fox-Powell, M. G., & Cousins, C. R. (2021). Partitioning of crystalline and amorphous phases during freezing of simulated Enceladus ocean fluids. *Journal of Geophysical Research: Planets*, *126*(1), e2020JE006628. https://doi.org/10.1029/2020JE006628

Glein, C. R., Postberg, F., & Vance, S. D. (2018). The geochemistry of Enceladus: Composition and controls. In P. M. Schenk, R. N. Clark, C. J. A. Howett, A. J. Verbiscer, & J. H. Waite (Eds.), *Enceladus and the icy moons of Saturn* (p. 39). University of Arizona Press. https://doi.org/10.2458/azu_uapress_9780816537075-ch003

Grima, C., Kofman, W., Hérique, A., & Beck, P. (2024). Revising the basal permittivity of the south polar layered deposits of Mars with a surficial dust cover. *Geophysical Research Letters*, *51*(12). https://doi.org/10.1029/2024gl109085

Gundlach, B., Ratte, J., Blum, J., Oesert, J., & Gorb, S. (2018). Sintering and sublimation of micrometre-sized water-ice particles: The formation of surface crusts on icy Solar System bodies. *Monthly Notices of the Royal Astronomical Society*, *479*(4), 5272–5287. https://doi.org/10.1093/mnras/sty1839

Hansen, C. J., Esposito, L., Stewart, A., Colwell, J., Hendrix, A., Pryor, W., et al. (2006). Enceladus' water vapor plume. *Science*, *311*(5766), 1422–1425. https://doi.org/10.1126/science.1121254

Hemingway, D. J., & Mittal, T. (2019). Enceladus's ice shell structure as a window on internal heat production. *Icarus*, *332*, 111–131. https://doi.org/10.1016/j.icarus.2019.03.011

Jabaud, B., Artoni, R., Tobie, G., Le Menn, E., & Richard, P. (2024). Cohesive properties of ice powders analogous to fresh plume deposits on Enceladus and Europa. *Icarus*, *409*, 115859. https://doi.org/10.1016/j.icarus.2023.115859

Kalousová, K., Schroeder, D. M., & Soderlund, K. M. (2017). Radar attenuation in Europa's ice shell: Obstacles and opportunities for constraining the shell thickness and its thermal structure. *Journal of Geophysical Research: Planets*, *122*(3), 524–545. https://doi.org/10.1002/2016je005110

Kempf, S., Beckmann, U., & Schmidt, J. (2010). How the Enceladus dust plume feeds Saturn's E ring. *Icarus*, *206*(2), 446–457. https://doi.org/10.1016/j.icarus.2009.09.016

Marion, G., Kargel, J., Catling, D., & Lunine, J. (2012). Modeling ammonia–ammonium aqueous chemistries in the Solar System's icy bodies. *Icarus*, *220*(2), 932–946. https://doi.org/10.1016/j.icarus.2012.06.016

Martin, E. S., Whitten, J. L., Kattenhorn, S. A., Collins, G. C., Southworth, B. S., Wiser, L. S., & Prindle, S. (2023). Measurements of regolith thicknesses on Enceladus: Uncovering the record of plume activity. *Icarus*, *392*, 115369. https://doi.org/10.1016/j.icarus.2022.115369

Matsuoka, K., MacGregor, J. A., & Pattyn, F. (2012). Predicting radar attenuation within the Antarctic ice sheet. *Earth and Planetary Science Letters*, *359*, 173–183. https://doi.org/10.1016/j.epsl.2012.10.018

Montagnat, M., & Duval, P. (2000). Rate controlling processes in the creep of polar ice, influence of grain boundary migration associated with recrystallization. *Earth and Planetary Science Letters*, *183*(1–2), 179–186. https://doi.org/10.1016/s0012-821x(00)00262-4

Moore, W. B., & Schubert, G. (2000). Note: The tidal response of Europa. *Icarus*, *147*, 317–319. https://doi.org/10.1006/icar.2000.6460

Nimmo, F., Bills, B., & Thomas, P. (2011). Geophysical implications of the long-wavelength topography of the Saturnian satellites. *Journal of Geophysical Research*, *116*(E11), E11001. https://doi.org/10.1029/2011je003835

Petrenko, V. F., & Whitworth, R. W. (1999). *Physics of ice*. OUP.

Phillips, R. J., Zuber, M. T., Smrekar, S. E., Mellon, M. T., Head, J. W., Tanaka, K. L., et al. (2008). Mars north polar deposits: Stratigraphy, age, and geodynamical response. *Science*, *320*(5880), 1182–1185. https://doi.org/10.1126/science.1157546

Plaut, J. J., Picardi, G., Safaeinili, A., Ivanov, A. B., Milkovich, S. M., Cicchetti, A., et al. (2007). Subsurface radar sounding of the south polar layered deposits of Mars. *Science*, *316*(5821), 92–95. https://doi.org/10.1126/science.1139672

Porco, C. C., Helfenstein, P., Thomas, P. C., Ingersoll, A. P., Wisdom, J., West, R., et al. (2006). Cassini observes the active south pole of Enceladus. *Science*, *311*(5766), 1393–1401. https://doi.org/10.1126/science.1123013

Postberg, F., Kempf, S., Schmidt, J., Brilliantov, N., Beinsen, A., Abel, B., et al. (2009). Sodium salts in E-ring ice grains from an ocean below the surface of Enceladus. *Nature*, *459*(7250), 1098–1101. https://doi.org/10.1038/nature08046

Scanlan, K. M., Young, D. A., & Blankenship, D. D. (2022). Non-linear radar response to the radial structure of Europa plume fallout deposits. *Icarus*, *378*, 114935. https://doi.org/10.1016/j.icarus.2022.114935

Schroeder, D. M., Bingham, R. G., Blankenship, D. D., Christianson, K., Eisen, O., Flowers, G. E., et al. (2020). Five decades of radioglaciology. *Annals of Glaciology*, *61*(81), 1–13. https://doi.org/10.1017/aog.2020.11



21699100, 2024, 12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024JE008615 by Dtsch Zentrum F. Luft-U. Raum Fahrt In D. Helmholtz Gemein., Wiley Online Library on [01/10/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Seiferlin, K., Kömle, N., Kargl, G., & Spohn, T. (1996). Line heat-source measurements of the thermal conductivity of porous $H_2O$ ice, $CO_2$ ice and mineral powders under space conditions. *Planetary and Space Science*, *44*(7), 691–704. https://doi.org/10.1016/0032-0633(96)00068-2

Souček, O., Běhounková, M., Schroeder, D. M., Wolfenbarger, N. S., Kalousová, K., Steinbrügge, G., & Soderlund, K. M. (2023). Radar attenuation in Enceladus' ice shell: Obstacles and opportunities for constraining shell thickness, chemistry, and thermal structure. *Journal of Geophysical Research: Planets*, *128*(2), e2022JE007626. https://doi.org/10.1029/2022je007626

Southworth, B. S., Kempf, S., & Spitale, J. (2019). Surface deposition of the Enceladus plume and the zenith angle of emissions. *Icarus*, *319*, 33–42. https://doi.org/10.1016/j.icarus.2018.08.024

Spencer, J., & Grinspoon, D. (2007). Inside Enceladus. *Nature*, *445*(7126), 376–377. https://doi.org/10.1038/445376b

Tajeddine, R., Soderlund, K. M., Thomas, P. C., Helfenstein, P., Hedman, M. M., Burns, J. A., & Schenk, P. M. (2017). True polar wander of Enceladus from topographic data. *Icarus*, *295*, 46–60. https://doi.org/10.1016/j.icarus.2017.04.019

Wolfenbarger, N. S., Carnahan, E., Jordan, J. S., & Hesse, M. A. (2021). A comprehensive dataset for the thermal conductivity of ice Ih for application to planetary ice shells. *Data in Brief*, *36*, 107079. https://doi.org/10.1016/j.dib.2021.107079

Wolfenbarger, N. S., Fox-Powell, M. G., Buffo, J., Soderlund, K. M., & Blankenship, D. D. (2022). Brine volume fraction as a habitability metric for Europa's ice shell. *Geophysical Research Letters*, *49*(22), e2022GL100586. https://doi.org/10.1029/2022gl100586



21699100, 2024, 12, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024JE008615 by Dtsch Zentrum F. Luft-U. Raum Fahrt In D. Helmholtz Gemein., Wiley Online Library on [01/10/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License